Chapter 8: Taxes

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March 2, 2026

Outline

Today we will talk about:

- Taxes
  - tax revenue
  - social welfare
  - tax incidence

Taxes

In the U.S. (Not necessarily in other countries!) direct price controls (ceilings/floors) are relatively rare.

Taxes are much more common: **What is a tax? Who is a tax levied on?**

When taxes are levied we are concerned with:

The **Incidence of the tax**: how the burden of the tax is shared among participants (buyers & sellers) in a market

How do we determine tax incidence?

Steps:

1. Determine whether the tax is on buyers or sellers
2. Decide which way the relevant curve shifts
3. Examine how the shifts affect equilibrium price and quantity
    1. Look at how buyers and sellers are affcected individually

Tax Revenue

How do we determine the tax revenue resulting from a tax?

$$\text{Tax Revenue } = (P_{Tax}^{Buyer} - P_{Tax}^{Seller}) \times Q_{Tax}^{*}$$
$$\text{Tax Revenue} = \text{Tax Wedge} \times Q_{Tax}^{*}$$

How can we determine incidence from this equation?

- Burden on buyers: $(P_{Tax}^{Buyer} - P^{*}) \times Q_{Tax}^{*}$
- Burden on sellers: $(P^{*} - P_{Tax}^{Seller}) \times Q_{Tax}^{*}$

Taxes

Let's return to the market for coffee...

Consider a \$0.5 tax on each cup of coffee *levied on the sellers:*

- Does this cause a shift in supply or demand?
- What will happen to P^* or Q^*?
- What is the tax revenue/where is it on a graph?
- What portion falls on buyers and what portion falls on sellers?

Taxes

In this example, buyers and sellers share the incidence of the tax.

Let's consider the same questions under the following extreme cases:

- Demand is perfectly inelastic
- Supply is perfectly inelastic

Taxes & Elasticity

- **Perfectly inelastic demand**: consumers buy the same amount of coffee no matter what $\implies$ consumers bear the entire incidence of the tax
- **Perfectly inelastic supply**: suppliers provide the same amount of coffee no matter what $\implies$ suppliers bear the entire incidence of the tax

Taxes & Elasticity

The relative elasticities determine the Tax Wedge:

This tells us which side of the market pays what portion of the tax.

In general:

- Demand is more elastic than supply $\rightarrow$ *suppliers* bear more of the burden
- Supply is more elastic than demand $\rightarrow$ *buyers* bear more of the burden

Really it is the **relative** elasticities which determine the wedge.

Taxes & Elasticity

And...

The dead weight loss resulting from a tax depends on elasticity of supply and demand:

- Demand or supply is elastic $\implies$ large dead weight loss
- Demand or supply is inelastic $\implies$ small deadweight loss

How do taxes and social welfare relate?

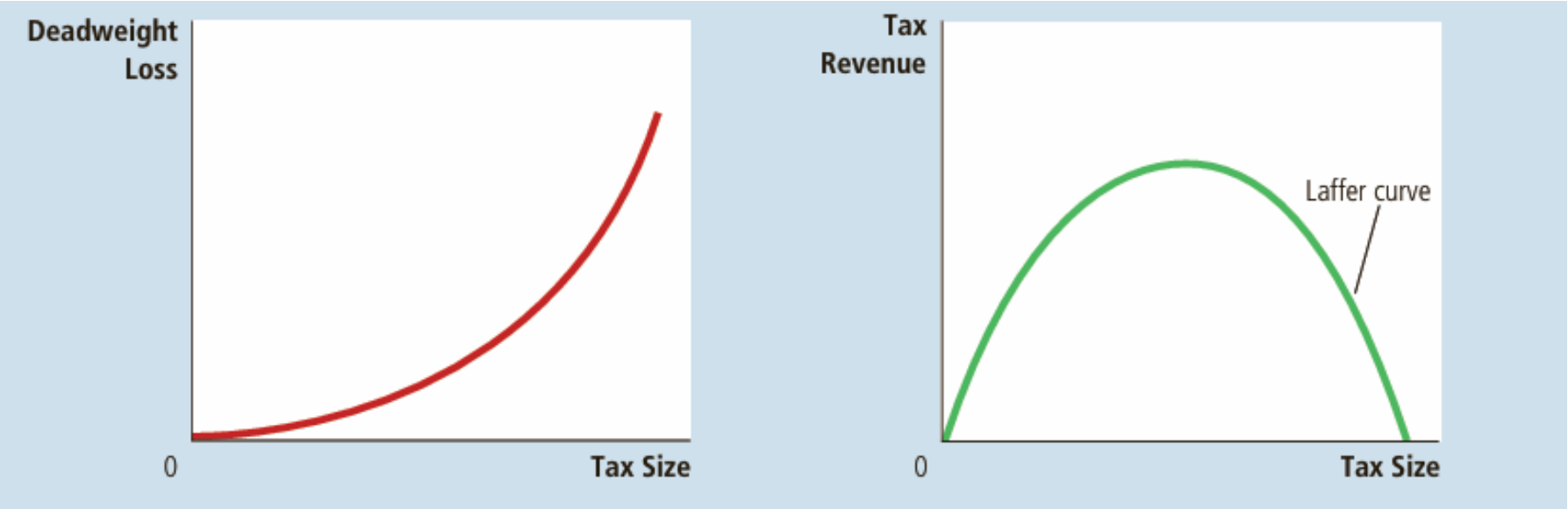

Figure 1: Dead weight loss and tax revenue

Logistics

Now I Will Distribute Exams
If Time, Consumer Theory!!!

Budget constraints

Most consumers want to increase the quantity/quality of the goods they consume, but money isn't infinite :(

Consumers face a **budget constraint:**

the limit on the consumption bundles that a consumer can afford

Example: A consumer would like to purchase pasta and wine. They have $100 to spend, pasta costs $20 per plate and wine costs $10 per glass...

1. How much of each good could they consume?
2. What is the equation for their budget constraint?
3. What does their budget constraint look like graphically?
4. What happens if the price of pasta decreases to $10?
5. What happens if income increases to $160?

Representing preferences

How do we know which bundle of goods a consumer prefers?

We can represent preferences using **indifference curves:**

IC: a curve showing consumption bundles that give the consumer *equal levels of satisfaction*

The slope at any point on an indifference curve equals the **Marginal Rate of Substitution (MRS)**: the rate at which a consumer is willing to give up one unit of good X for another unit of good Y

We calculate MRS as follows:

$$MRS = \frac{MU_X}{MU_Y}$$

Properties of indifference curves:

1. Higher indifference curves are preferred to lower ones:
    1. More is better!
2. Indifference curves are downward sloping:
    1. in most cases a consumer likes both goods, so to decrease plates of pasta but keep the same satisfaction, we must increase glasses of wine
3. Indifference curves do not cross
    1. If they crossed, we would violate the assumption that people prefer more to less
4. Indifference curves are bowed inwards
    1. MRS is not constant... we are more willing to give up a good we already have an abundance of

Optimizing

A consumer wants to maximize their satisfaction given their budget constraint...

This is achieved by choosing the bundle where either:

- Their indifference curve is tangent to their budget constraint (where they touch... kind of)
- The MRS is equal to the slope of the budget constraint

When income or prices of goods change, our budget constraint does too, and so we would find ourselves on a new indifference curve at optimum...

Income & Substitution Effects

When the price of goods changes, the resulting change to the budget constraint and optimal bundle is a result of two effects:

1. **Income effect:** Change in consumption that results when a price change *moves the consumer to a higher or lower indifference curve*
2. **Substitution effect:** Change in consumption that results when a price change *moves the consumer along a given indifference curve to a point with a new MRS*

** A change in income results only in an income effect **